

Method	Pick	Look	Clean	Heat	Cool	Pick2	All	Score	Succ.
<i>Closed-source LLMs</i>									
GPT-4o	75.3	60.8	31.2	56.7	21.6	49.8	48.0	31.8	23.7
Gemini-2.5-Pro	92.8	63.3	62.1	69.0	26.6	58.7	60.3	42.5	35.9
<i>Qwen2.5-7B-Instruct</i>									
Qwen2.5	33.4	21.6	19.3	6.90	2.80	3.20	14.8	26.4	7.80
<i>Prompt-based Agentic or Memory-based Methods</i>									
ReAct*	48.5	35.4	34.3	13.2	18.2	17.6	31.2	46.2	19.5
Reflexion*	62.0	41.6	44.9	30.9	36.3	23.8	42.7	58.1	28.8
Mem0	54.0	55.0	26.9	36.4	20.8	7.69	33.6	23.9	2.00
ExpeL	21.0	67.0	55.0	52.0	71.0	6.00	46.3	30.9	11.2
MemP	54.3	38.5	48.1	56.2	32.0	16.7	41.4	25.3	6.40
SimpleMem	64.5	33.3	20.0	12.5	33.3	3.84	29.7	33.2	8.59
<i>RL-based Methods</i>									
RLOO*	87.6	78.2	87.3	81.3	71.9	48.9	75.5	80.3	65.7
GRPO*	90.8	66.1	89.3	74.7	72.5	64.7	77.6	79.3	66.1
<i>Memory-Augmented RL-based Methods</i>									
MemRL	62.8	38.5	22.2	12.5	8.00	0.00	21.4	29.5	9.20
EvolveR	64.9	33.3	46.4	13.3	33.3	33.3	43.8	42.5	17.6
Mem0+GRPO	78.1	54.8	56.1	31.0	65.0	26.9	54.7	58.1	37.5
SimpleMem+GRPO	89.5	36.3	60.0	50.0	64.9	26.3	62.5	67.8	46.9
SKILLRL	97.9	71.4	90.0	90.0	95.5	87.5	89.9	85.2	72.7